Attention-based Imputation of Missing Values in Electronic Health Records Tabular Data

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***Abstract*—The imputation of missing values (IMV) in electronic health records tabular data is crucial to enable machine learning for patient-specific predictive modeling. While IMV methods are developed in biostatistics and recently in machine learning, deep learning-based solutions have shown limited success in learning tabular data. This paper proposes a novel attention-based missing value imputation framework that learns to reconstruct data with missing values leveraging between-feature (self-attention) or between-sample attentions. We adopt data manipulation methods used in contrastive learning to improve the generalization of the trained imputation model. The proposed self-attention imputation method outperforms state-of-the-art statistical and machine learning-based (decision-tree) imputation methods, reducing the normalized root mean squared error by 18.4% to 74.7% on five tabular data sets and 52.6% to 82.6% on two electronic health records data sets. The proposed attention-based missing value imputation method shows superior performance across a wide range of missingness (10% to 50%) when the values are missing completely at random.**

***Index Terms*—missing value imputation, self-attention, between-sample attention, tabular data, electronic health records.**

I. Introduction

The machine learning literature assumes the availability of clean and curated datasets for benchmarking algorithms. However, such an assumption does not hold for electronic health records (EHR) tabular data because physicians may not recommend specific lab tests during patient follow-ups. Therefore, many lab values go unrecorded and are missing in EHR tables. Unfortunately, advanced machine learning solutions are not operable on data with missing values. A large volume of EHR data remains unprepared for machine learning due to missing values, which could have significantly contributed to health science and patient risk prognosis [1]. The imputation of missing values (IMV) enables machine learning and ensures the inclusion of patient samples and potentially important clinical variables that are otherwise discarded due to missingness. Despite the importance of IMV, model-generated imputation values can negatively impact data quality and data-driven outcomes. Therefore, the accuracy of IMV in EHR tabular data is imperative in reaping the benefits of machine learning and reducing misdiagnoses, which may otherwise lead to improper therapies or missed chances for timely interventions.

The study of missing values is traditionally investigated in biostatistics, which has laid the foundation of recent advancements in machine and deep learning-based imputation methods. Advanced machine learning methods for IMV use the multiple imputations framework, such as multiple imputations by chained equations (MICE) [2]. The chains of linear regression equations are replaced by machine learning regressors, such as random forest, gradient boosting, and deep regressors. On the other hand, deep learning-based IMV solutions have been proposed based on generative learning (e.g., generative adversarial networks (GAN) [3], [4] or self-supervised learning (e.g., autoencoder) [5]. Conversely, attention-based learning sees limited presence in learning tabular data, let alone in missing value imputation, despite its groundbreaking achievements in image and text data learning. This paper introduces a novel attention-based method for imputing missing values in EHR tabular data by learning attention between patient samples or attention between laboratory variables. We hypothesize that attending between EHR variables via attention models would yield better estimates of missing values than state-of-the-art regression models within multiple imputations framework.

The rest of the paper is presented as follows. Section 2 presents an overview of state-of-the-art missing values imputation methods. Section 3 details the proposed algorithm and experiments conducted in our study. In Section 4, we present the imputation accuracy of our proposed method and compare it against other state-of-the-art baseline IMV methods. Section 5 summarizes the key findings, offers insights into the results, and suggests potential directions for future research to conclude the paper.

II. Background

The presence of missing values in healthcare, specifically in EHR tabular data, is a significant barrier to machine learning of patient outcomes. Imputation methods are imperative to

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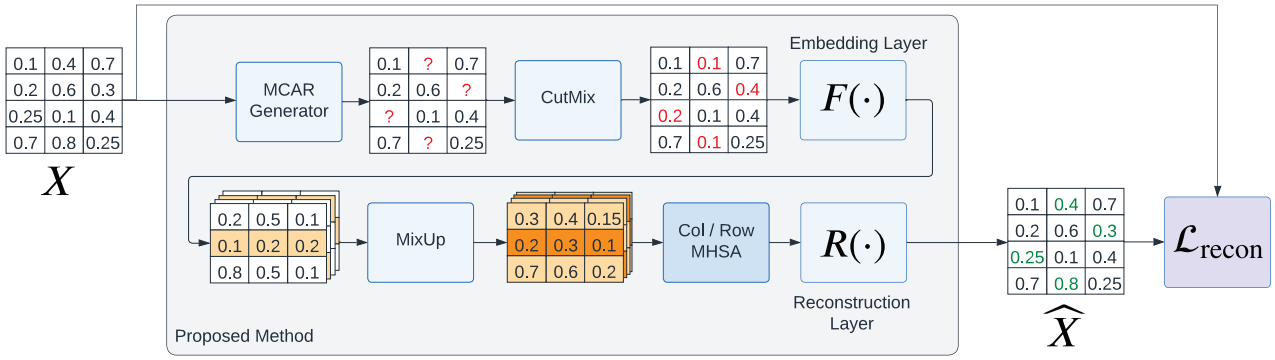


Fig. 1. Missingness is simulated using a missing completely at random (MCAR) mask, which is initially filled via CutMix [6]. The initially filled data are used to generate feature embeddings via an embedding layer (Equation 2) before applying MixUp [7]. The result is input to a multi-head attention-based network for column (between-feature) or row (between-sample) attention modeling. The output is reconstructed back to original input (Equation 5). This network uses multi-headed self-attention (MHSA) [8] that performs attention between features (Equation 3) or between samples (Equation 4).

complete a dataset for machine learning-based prognosis and decision-making in healthcare.

A diverse set of machine learning (ML) and statistical models have been proposed for imputing missing values in tabular data. The multiple imputations (MI) is a popular statistical framework that has inspired the development of multiple imputations using chained equations (MICE) [9], missForest [10], and deep MICE [2]. MICE iteratively updates a chain of regression equations (e.g., linear regression) modeled to estimate each variable as a function of other variables. While the conventional MICE model uses linear regression equations, replacing these equations with non-linear machine learning regressors (e.g., random forest and gradient-boosting trees) has shown promising improvements in imputation accuracy. However, Mera et al. argue that MICE is computationally expensive when working with large datasets or complex imputation models [11]. The number of regression equations grows with the number of variables, which makes it computationally expensive. missForest is an iterative imputation-based method that can handle large and complex data sets, achieving better imputation accuracy than MICE [10]. Liao et al. argue that missForest is not suitable for datasets with high missing values for being prone to overfitting [12]. There is more research on missing value imputations that uses an ensemble of machine learning models, such as gradient boosting and random forest [13], [14]. However, traditional ML methods are not structured to learn feature representations in cognizant of a learning objective, unlike deep learning methods [15].

Despite the tremendous success of deep learning methods in other data domains (e.g., computer vision, signal processing, natural language processing), it has not seen much success and acceptance in learning tabular data. Traditional machine learning methods still dominate tabular data as the literature suggests that deep learning is the ”last unconquered castle” for tabular data [16].

A deep neural network with the MICE framework outperforms traditional MICE in all experimental scenarios up to 80% missing values [2]. Generative deep learning models are popular where a diffusion-based framework (MissDiff) is trained to generate values for missing data entries [17]. Apart from these studies, attention and transformer-based learning algorithms proposed for tabular data, such as Feature Tokenizer Transformer (FT-Transformer) [18] and Non-parametric Transformer (NPT) [19], outperform traditional ML models in multiple tabular data classification tasks. Similarly, Self-Attention and Intersample Attention Transformer (SAINT) learn between features and between sample attention in addition to using contrastive learning at the pretraining stage [20]. However, these state-of-the-art deep learning solutions for classifying tabular data have not been considered for solving missing value imputation problems.

A recent study proposes a denoising self-attention network (DSAN) to impute missing values in numerical and category columns [21]. In another recent work, an imputation method learns attention between features using the observed part of the data matrix, excluding the updates in missing value estimation [22]. Therefore, data sets with high missing rates would not have sufficient values to compute between feature attention in this method. Furthermore, experimental results show that tree-based imputation methods (e.g., missForest) outperform their attention-based method on tabular data sets with only numerical values.

The gaps in the literature inspire us to propose a novel attention-based missing value imputation method that can incorporate missing value estimates in computing between-feature (self-attention) or between-sample attentions. The imputation accuracy of the proposed attention-based method is compared against other state-of-the-art missing value imputation methods for varying missing rates.

III. Methods

This section presents the proposed method in detail, as illustrated in Figure 1. The proposed method is then evaluated against other statistical and machine learning based imputation methods.

A. Data sets

We use five datasets from the OpenML [23] tabular data repository for our experiments, as demonstrated in Table I.

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TABLE I
SUMMARY OF TABULAR DATASETS USED IN THIS STUDY. THE IMPUTATION METHODS ARE FIRST EVALUATED ON REGULAR TABULAR DATA, SHOWN WITH OPENML ID, WHICH ARE THEN TRAINED AND TESTED ON TWO ELECTRONIC HEALTH RECORDS (EHR) DATA SETS.

ID	Dataset Name	Samples	Features	Classes
1494	Qsar biodeg	1055	41	2
1497	Wall robot navigation	5456	24	4
1510	Breast cancer	569	30	2
4538	Gesture phase	9873	32	5
40994	Climate model	540	20	2
EHR	Sepsis	2164	43	2
EHR	Hypotension	3910	19	4

Tabular data sets are selected from diverse domains with varying sample and feature sizes. Following this initial validation, we evaluated the imputation method on two publicly available electronic health records (EHR) tabular data sets: 1) Sepsis and 2) Hypotension. Both datasets are sourced and validated from the MIMIC-III database by the Health Gym project [24]. The Sepsis data set records 43 clinical variables of 2,164 patients tracked in 20 follow-ups. Similarly, the Hypotension data set includes 3,910 patients recorded for 19 clinical variables over 20 follow-ups. In this paper, we use last observed (or follow-up) measurements for our experiments instead of longitudinal patient follow-ups [25].

B. Attention-based feature reconstruction

The Transformer architecture [8] with residual learning [26] and the dot-scaled attention has been adapted for tabular data [18]–[20], [27]. The section explains the main parts of the network.

1) Feature embeddings: A tabular data set with b samples and n features with shape (b, n) is transformed to a d-dimensional feature embedding, F, of shape (b, n, d) [18]–[20]. The j-th feature is transformed into a d-dimensional embedding using a linear layer with learnable weight W_j and bias B_j parameters. For categorical features, a lookup table or one-hot encoding vector (e_j) is used to select different weight matrices for each value while using the same bias matrix. This transformation is shown in equation 1. Next, n number of d-dimension vectors of i-th sample are stacked vertically to create feature embedding F^i of shape (n, d), which is aggregated across b number of samples to produce F of shape (b, n, d). This process is shown in equation 2.

$$f_j^i = \begin{cases} W_j x_j + B_j, & \text{if } x_j \text{ is numerical} \\ (e_j)^T W_j^{(\text{cat})} + B_j, & \text{if } x_j \text{ is categorical} \end{cases} \quad (1)$$

$$F(X) = [F^1, F^2, ..., F^b] \quad \text{where } F^i = \text{stack } [f_1^i, f_2^i, ..., f_j^i] \quad (2)$$

2) Attention between features: The feature embedding F is updated to G to capture between-feature attention (self-attention) using dot-scaled attention [8]. Vaswani et al. computes attention using multiple heads on small feature embeddings. They refer to this as multi-headed self-attention (MHSA), as summarized in equation 3.

$$G^i = \text{MHSA}(F^i) = \text{ concat}(\text{head}_1, ..., \text{head}_H)W^C, \quad (3)$$
$$\text{where head}_h = \text{softmax}(\frac{Q_h(K_h)^T}{\sqrt{d_H}})V_h, and$$
$$Q_h, K_h, V_h = F^i W_h^Q,\ F^i W_h^K,\ F^i W_h^V$$

Here, Equation 3 shows MHSA concatenating H-heads where each head outputs an embedding of dimension $d_H = \frac{d}{H}$. W_h^Q, W_h^K, and W_h^V are the learnable weight parameters of shape (d, d_H) for the h-th attention head. F_i is defined in Equation 2. Outputs from each attention head (n, d_H) are concatenated to shape (n, d) and projected to G^i via a linear layer that uses a learnable weight W^C of shape (d, d).

3) Attention between samples: F is reshaped to $F'(b \times n, d)$ following [19], [20]. Each of the $b \times n$ rows in F' contains a d-dimensional embedding, which is input to MHSA (Equation 3). The output of MHSA (H') is reshaped back to $H(b, n, d)$ as below before feature reconstruction via the subnetwork.

$$H \leftarrow \text{reshape}(H') \leftarrow H' \leftarrow \text{MHSA}(F') \leftarrow \text{reshape}(F) \quad (4)$$

4) Feature reconstruction: A stack of multi-layered perception (MLP) heads is used to reconstruct the original feature values from their final feature embeddings G (between-feature or self-attention) or H (between sample attention). An MLP head decodes each d-dimensional vector to the j-th feature, $\hat{X}_j$, as shown in Equation 5. The reconstruction loss $(\mathcal{L}_{recon})$ is computed using mean squared error $(\mathcal{L}_{num})$ for numerical features and cross-entropy $(\mathcal{L}_{cat})$ for categorical features, as shown below.

$$R(F_j) = \text{MLP}_j(F_j) = \hat{X}_j \quad (5)$$

$$\mathcal{L}_{num}(\hat{X}_j, X_j) = \frac{1}{b}\sum_{i=1}^{b}(\hat{x}_j^i - x_j^i)^2, \quad (6)$$

$$\mathcal{L}_{cat}(\hat{X}_j, X_j) = -\frac{1}{b}\sum_{i=1}^{b}\sum_{c=1}^{C_j} x_c^i \log(\hat{x}_c^i), \quad (7)$$

$$\mathcal{L}_{recon} = \sum_{j=1}^{n} \begin{cases} \mathcal{L}_{cat}(\hat{X}_j, X_j) & \text{if } X_j \text{ is categorical} \\ \mathcal{L}_{num}(\hat{X}_j, X_j) & \text{otherwise} \end{cases} \quad (8)$$

Here, $\hat{X}_j$ and X_j represent the reconstructed and original values of the j-th feature, respectively. If X_j is categorical, x_c^i is the binary label indicating whether the i-th sample includes the c-th category, with C_j representing the total number of categories for the j-th categorical feature.

C. Data manipulation in attention learning

We simulate MCAR-type missingness at every mini-batch while training the attention-based imputation method. This allows the model to learn different patterns of MCAR-type missingness on the training data folds. We adopt the data manipulation concepts of CutMix [6] and MixUp [7] from contrastive learning. The training data folds with missing values are then processed in two steps: initial values for

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TABLE II
NORMALIZED ROOT MEAN SQUARE ERROR (NRMSE) OBTAINED ON OPENML [23] DATA SETS. BETWEEN-FEATURE AND BETWEEN-SAMPLE ARE THE PROPOSED ATTENTION-BASED IMPUTATION METHODS.

OpenML ID	Missing Rate (%)	Between Feature	Between Sample	missForest [10]	MICE [9]	Median
1494	10	**0.59 (0.1)**	0.66 (0.1)	1.17 (0.3)	0.91 (0.2)	2.03 (0.2)
	30	**0.77 (0.1)**	0.80 (0.1)	2.07 (0.4)	2.21 (0.5)	3.76 (0.3)
	50	**0.88 (0.1)**	0.92 (0.1)	3.35 (0.3)	3.54 (0.4)	4.88 (0.3)
1497	10	**0.21 (0.0)**	0.24 (0.0)	0.61 (0.0)	0.77 (0.0)	1.01 (0.0)
	30	**0.34 (0.0)**	0.37 (0.0)	1.19 (0.0)	1.60 (0.0)	1.75 (0.0)
	50	**0.45 (0.0)**	0.47 (0.0)	1.78 (0.0)	1.99 (0.0)	2.27 (0.0)
1510	10	0.79 (0.2)	1.16 (0.2)	0.62 (0.4)	**0.24 (0.1)**	3.05 (1.1)
	30	1.19 (0.3)	1.61 (0.3)	1.37 (0.6)	**0.96 (0.3)**	5.86 (0.9)
	50	**1.72 (0.2)**	1.73 (0.3)	2.37 (0.6)	2.11 (0.5)	7.42 (1.3)
4538	10	**1.90 (0.1)**	3.19 (0.1)	3.90 (0.4)	4.27 (0.4)	9.30 (0.2)
	30	**2.93 (0.1)**	4.41 (0.1)	9.39 (0.5)	10.14 (0.5)	16.04 (0.3)
	50	**4.04 (0.1)**	4.82 (0.0)	14.63 (0.8)	15.47 (0.3)	20.55 (0.3)
40994	10	**0.25 (0.0)**	0.34 (0.0)	0.82 (0.0)	0.76 (0.1)	0.76 (0.1)
	30	**0.41 (0.0)**	0.44 (0.0)	1.49 (0.0)	1.34 (0.0)	1.34 (0.0)
	50	**0.50 (0.0)**	0.52 (0.0)	2.06 (0.0)	2.10 (0.2)	1.73 (0.0)

missing input data are obtained via CutMix, followed by another data manipuation step using MixUp in the embedding space.

1) Missingness generation: A data table X with missing values is composed of X_{obs} and X_{mis}, where X_{obs} is the observed part of X and X_{mis} presents the missing entities. A mask matrix M_j^i includes 1 if the corresponding element in X is observed and 0 if the value is missing. This study adheres to the missing completely at random (MCAR) type of missingness, where the likelihood of any data point M_j^i being missing is independent of the observed data X_{obs} or missing data X_{mis}, as shown in Equation 9.

$$P(M = 0) = P(M = 0 \mid X_{\text{obs}}, X_{\text{mis}}) \tag{9}$$

A data table X is passed to the MCAR generator to simulate missingness with one of the three missing rates: 10%, 30%, and 50%.

2) Initializing missing values: A data set with missing values must be completed using some initial values before training a deep learning model. We propose two data manipulation techniques, CutMix and MixUp at each minibatch while training our attention model. We use the CutMix method to replace masked values (M^i) of each sample (x^i) using corresponding feature values of another random sample, as shown in Equation 10. Next, the imputed data set is used to generate the embedding, on which we apply the second data manipulation method, MixUp [7]. MixUp corrupts the sample embeddings by replacing it with a linear combination of embeddings of the current F^i and another sample selected at random F^j, as shown in Equation 11.

$$\tilde{x}^i = x^i \odot M^i + x^j \odot (1 - M^i) \tag{10}$$

$$\tilde{F}^i = \alpha \cdot F^i + (1 - \alpha) \cdot F^j \tag{11}$$

Here, α denotes the MixUp strength.

D. Experimental setup

We use a five-fold cross-validation scheme where each test fold (20% data) is imputed using an imputation model trained by four training folds (80%). A fixed random state seed reproduces the same train-test folds in all experiments.

We repeat all experiments for three missing rates (10%, 30%, and 50%). The proposed attention network is trained using a mini-batch gradient descent with a batch size of 128, and Adam optimizer with β_1 = 0.9, β_2 = 0.999, decay = 0.01, and a learning rate of 0.0001. The embedding size is set to 32. Eight attention heads are chosen with transformer depth (sequential MHSA blocks) equal to 6, including a dropout rate of 0.1 and MixUp strength α equals 0.2. We train the attention network for 1000 epochs and take the trained model for reconstructing or imputing missing values on test data folds.

E. Model evaluation

The imputation score of the proposed attention network is compared against several state-of-the-art baseline imputation methods, including MICE [9], missForest [10], and median imputation.

The error between the imputed and ground truth values in the test data fold is measured by taking the root mean squared error (RMSE) as shown in Equation 12. The RMSE is then divided by the mean of the original values to calculate the normalized root mean squared error (NRMSE), as shown in Equation 13. This normalization shows the imputation error relative to the scale of the variable.

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(X_i - \hat{X}_i)^2}{N}} \tag{12}$$

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TABLE III
NORMALIZED ROOT MEAN SQUARED ERROR (NRMSE) ON ELECTRONIC HEALTH RECORDS (EHR) DATA SETS OBTAINED FROM THE MIMIC-III [24] REPOSITORY WITH A MISSING RATE OF 30%. BETWEEN-FEATURE AND BETWEEN-SAMPLE ARE THE PROPOSED ATTENTION-BASED IMPUTATION METHODS.

EHR Data	Between Feature	Between Sample	missForest [10]	MICE [9]	Median
Sepsis	**1.77 (0.1)**	2.40 (0.1)	4.03 (0.3)	3.74 (0.3)	7.00 (0.4)
Hypotension	**1.55 (0.0)**	2.23 (0.1)	8.91 (0.8)	8.91 (0.6)	10.89 (0.7)

$$\text{NRMSE} = \frac{\text{RMSE}}{\frac{1}{N}\sum_{i=1}^{N} X_i} \tag{13}$$

IV. RESULTS

This section presents the performance of the proposed attention-based imputation method. All the experiments are conducted on a Ubuntu 22.04 machine with Intel(R) Xeon(R) W-2265 CPU @ 3.70GHz CPU with 64GB RAM and Quadro RTX A4000 16GB GPU. The processor has 24 logical cores. We use deep learning methods and modules available in the PyTorch package. PyTorch automatically utilizes multiple CPUs with parallel processing.

A. Imputation of general tabular data

Table II shows that missForest outperforms MICE on three of five OpenML tabular data sets. As expected, median imputation registers the highest (worst) NRMSE among all imputation methods. missForest generally achieves a better (lower) NRMSE than MICE at higher missing rates, except for data set ID 1510.

The proposed attention-based method, either between-feature or between-sample attention, yields substantial improvement in imputation accuracy compared to other baselines, as shown in Table II. The only exception is the data set ID 1510 when MICE is the best and missForest is the second best methods for up to 30% of missingness. Our proposed attention-based imputation yields substantially better imputation accuracy compared to the second based imputation method, missForest, as shown in Figure 2.

It is noteworthy that the current attention-based imputation, specifically AimNet [22], is outperformed by missForest on datasets predominantly featuring numerical data. The method is tested on data with 20% missing values only. In contrast, our proposed between-feature and between-sample attention methods outperform missForest for a wide range of missing rates (10% to 50%).

B. Imputation of EHR tabular data

The performance of the proposed attention-based imputation on tabular data is subsequently evaluated on two available electronic health records datasets: Sepsis and Hypotension at 30% missingness, as shown in Table III.

For the Sepsis data set, the proposed between-feature attention yields 56% and 53% improvement in NRMSE over missForest and MICE, respectively. Similarly, between-sample attention shows 40% and 35% improvements in NRMSE over missForest and MICE, respectively.

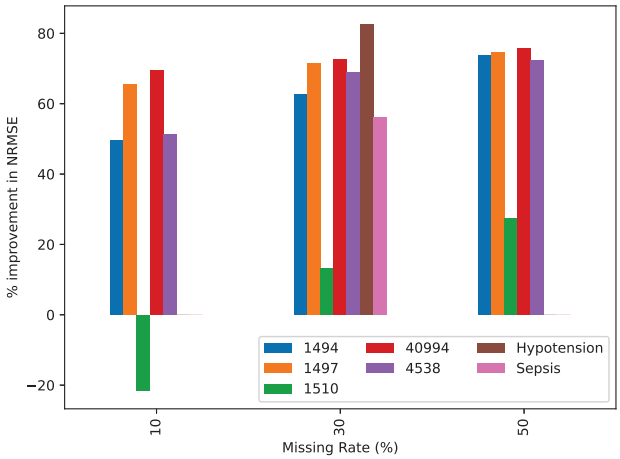


Fig. 2. Percentage improvement in normalized root mean squared error (NRMSE) achieved by the proposed attention based imputation method compared to the second best missForest imputation. Results are shown on five openML tabular data sets with IDs and two health records data sets (Hypotension and Sepsis).

For the Hypotension dataset, the proposed attention-based learning outperforms statistical and machine learning baseline methods. Specifically, between-feature attention yields 83% better NRMSE than missForest and MICE. Likewise, between-sample attention improves the NRMSE by 75% against missForest and MICE.

V. CONCLUSIONS

The paper introduces a novel attention-based learning framework for missing value imputation, incorporating data manipulation techniques from contrastive learning. The imputation results on separate test data with missing values via a five-fold cross-validation scheme reveal the superiority of the proposed method over other baselines by a large margin and across a range of missing rates. The proposed imputation method generalizes well across standard tabular datasets and those found in electronic health records. In the future, more rigorous experiments will be conducted on other missingness types and missing rates.

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